The Monte Carlo Method, an Introduction

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1 What is a Monte Carlo calculation?

In an early lecture note (around 1960, but see also [1]) one of the pioneers of the Monte Carlo technique M.H. Kalos, quotes the two "definitions": 1) a last resort when doing numerical integration and 2) a way of wastefully using computer time. Today common assumptions characterize it as a numerical method involving random numbers in a significant way.

In a certain sense any large computer calculation has random aspects, due to roundoff errors. Also deterministic "molecular dynamics" (MD) calculations, in which the interaction of a large number of moving particles is followed by integrating Newton's equation can have random results, due to randomly chosen initial conditions and/or due to the large number of particles. These are usually excluded from that definition of Monte Carlo techniques. The involvement of randomness in Monte Carlo methods is rendered more precisely to mean *deliberate use of random numbers in a calculation which has the structure of a stochastic process.*

Two major areas of application are in statistical mechanics (many particle systems) and in linear kinetic (particle) transport theory. The first type of calculations are embodied in a very specific sampling technique, and are not discussed in this lecture. The second, for example traffic flow, finance, genetics, but in particular neutronics, radiation transport, cosmic rays, neutral and charged particle transport in plasmas, etc., rely on a study of many interesting stochastic processes by imitating the random processes directly on the computer. Although intuition, and the resulting high transparency of the procedure, is an important ingredient in this type of stochastic analysis, a sound mathematical basis also exists. This allows rigorous mathematical proofs to

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be given that certain methods actually provide "solutions" to certain generic mathematical equations, e.g., to Fredholm Integral Equations of second kind in case of transport problems. Distinct from most numerical schemes, in these stochastic methods error estimates are provided by the method itself rather than requiring additional considerations. Some mathematical background and basic statistical results are also needed to analyze results of Monte Carlo simulations, for estimation of errors and for obtaining more economical approaches beyond simple simulation of nature.

There exists a vast amount of introductory literature, from a basic textbook level up to monographs focussing on very specialized applications (see, for example, [1, 2, 3, 4]), probably now hundreds of web-based lecture notes and uncounted journal articles. This present introductory chapter certainly duplicates most, if not all of that material. Our aims are to introduce the terminology, and to convey the message that Monte Carlo Methods do have a solid basis in measure theory (with the theory of probability as special case thereof). Strict mathematical proofs of convergence of the method to the exact solution exist, but also, and distinct from most numerical concepts, implementation be strongly guided by intuition and retain a high transparency even in very complex situations.

We will, after some short historical remarks below, start with introducing the concepts of random events, of error estimates and unbiased procedures for estimation.

Practical implementations of Monte Carlo techniques rely on our ability to draw random number from any probability law we wish. Only a few, most basic facts and concepts in this regard will be repeated here in the second section.

In the third section the central limit theorem and variance reduction techniques will be demonstrated "at work" using the generic example for Monte Carlo methods: integration by stochastic sampling. The relation of this very general but intuitively and transparent application to the mathematically and statistically more involved transport problems, ("Monte Carlo particle simulation") will be frequently used as guidance here and will be discussed in more detail in a separate chapter of this lecture notes [5].

1.1 Historical notes

Monte Carlo concepts fall into the branch of experimental mathematics. In ordinary mathematics conclusions are deduced from postulates (Deduction). In experimental mathematics conclusions are inferred from observations (Induction). Monte Carlo methods comprise that branch of experimental mathematics, which is concerned with experiments on random events (mainly: random numbers). Monte Carlo methods can be of probabilistic or deterministic type.

Usually the first reference to the Monte Carlo Method is the famous needle experiment of Comte de Buffon (1733), a French biologist (1707-1788), Figure 1. Buffon pointed out that if a needle of length L is tossed on a plane with

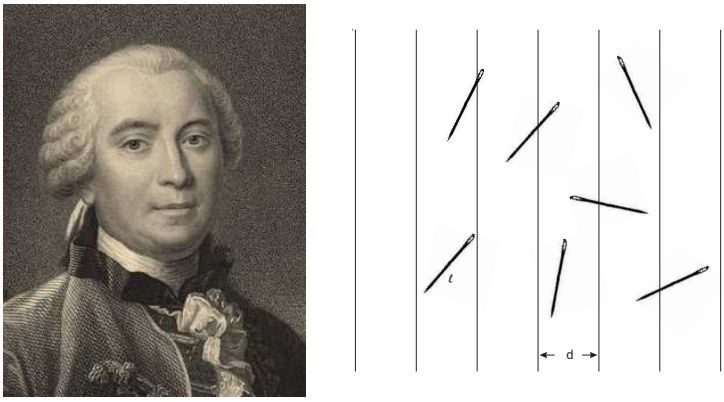


Fig. 1. Buffon's needles: What is the probability p, that a needle (length L), which falls randomly on a sheet, crosses one of the lines (distance D)

parallel lines a distance D apart ($D > L$), it has probability $p = \frac{2L}{\pi D}$ to fall such that is crosses one of the lines. Later, also Laplace suggested this procedure to determine π by counting the number of crosses n in N repetitions of the experiment. Then:

$$\frac{n}{N} = \frac{2L}{\pi D} \Rightarrow \pi \approx \frac{2L}{D} \cdot \frac{N}{n} \ . \tag{1}$$

This historical use of a "Monte Carlo procedure" has all key features of the method:

- Convergence: about N=100000 trials are needed for only 2 digits after the comma. Convergence is slow, but foolproof.
- Transparency: The method is intuitively understandable, even without any mathematical reasoning.
- error estimates, optimization: Error estimates and optimal choice of L, D are provided by theory of probability (Binomial distribution, statistical variance as 2nd central moment etc.).

Modern use Monte Carlo techniques, in the age of digital computers, was initiated by the pioneering work of John von Neumann and Stanislaw Ulam in thermonuclear weapon development. They are also credited for having coined the phrase "Monte Carlo".

Many monographs on Monte Carlo Methods start with an introduction to measure theory and in particular to elementary probability theory. Although we will introduce and use the proper mathematical vocabulary too, we will, with respect to purely mathematical aspects, refer to those and largely rely upon the intuitive meaning.

We refer in particular to the classic monograph by Hammersley and Handscomb, [2]. This book provides a short and very readable overview of Monte Carlo. Remarkably, the theoretical foundations today remain rather

similar to those from 1964, when this book was first published. Just the applications are far more sophisticated today. The illustrative examples on Monte Carlo integration and some of the advanced techniques in this present introduction will be based upon this text. (A pdf-file of that book, which is out of print since long, can be downloaded from the web, e.g.: http://www.eirene.de/Hammersley-Handscomb.pdf)

1.2 The basic principle

The principle is to find (estimate) mean values, i.e. expectation values, I of some system components. If a deterministic problem is to be solved (e.g., a definite integral, a Matrix inversion,...) one first has to invent a stochastic system (Ω, σ, p, X) such that a mean value (= expectation value $E(X)$) of that system coincides with the desired solution I of the deterministic problem. In other cases, if the problem is defined already in probabilistic terms, then this can directly be taken as a basis for an "analogue" simulation and estimation.

In any case: I is a single numerical quantity of interest (not an entire functional dependence), and one might always think of I as some definite integral.

The simple intuitive interpretations are given below, but in abstract mathematical terms this stochastic model is given by the *probability space* (Ω, σ, p, X). Ω is a set of elementary (random) events ω, the σ-field is a set of subsets of Ω to which the measurable function p assigns a value (the "probability") from the interval [0, 1], such that the Kolmogoroff axioms for a "probability" are fulfilled. X is a random variable on Ω, assigning a (usually real) number (or vector) to each random event, e.g.: $X(\omega) \to \mathbb{R}$, such that $I = E(X)$, the expected value of X.

The expectation value $E(X)$ and variance $\sigma^2(X)$ are defined as the first moment and second central moment, respectively, and, unless otherwise stated, we assume that they both exist:

$$E(X) := \int_\Omega dp \;\; X \quad \text{and} \quad \sigma^2(X) := \int_\Omega dp \;\; (X - E(X))^2 \tag{2}$$

(Note: $E(X) = E_p(X)$, $\sigma^2(X) = \sigma_p^2(X)$, i.e., the moments of X of course depend upon the probability measure p.)

A stochastic approximation to I is then obtained by producing an independent sequence of random events $\omega = [\omega_i, i = 1, \ldots N]$ according to probability law p and evaluating

$$E(X_N) = I_N(\omega) = \frac{1}{N} \sum_{i=1}^{N} X(\omega_i) \; . \tag{3}$$

The "estimator" I_N is just the arithmetic mean of many (N) evaluations $X(\omega_i)$ of the repeated random experiment.

Even without any of this abstract mathematical background it is intuitively clear (see examples below) that I_N will converge, in some sense, to $E(X)$, hence to I by construction, as the number of samples N is increased. However the "laws of large numbers" and the "central limit theorems" of probability theory not only provide sound mathematical proofs that this Monte Carlo procedure is exact ("unbiased") but also that it converges: $I_N \to I$ for $N \to \infty$, albeit slowly (with $1/\sqrt{N}$). In particular the "central limit theorem of probability theory" (see any textbook on Monte Carlo, or Probability Theory) asserts that the distribution of the random variable $I_N(\omega)$, for large enough N, converges to a Gaussian distribution, with mean value $I = E(X)$ and variance $\sigma^2(I_N) = \sigma^2(X)/N$. Hence the typical results from statistical error analysis under Gaussian distribution laws apply, e.g., also the resulting confidence levels. It is, therefore, common practice in Monte Carlo applications to quote results as

$$I \approx I_N \pm \sigma(I_N), \quad \text{or:} \quad I \approx I_N \pm 2 \cdot \sigma(I_N), \tag{4}$$

which have confidence levels of about 66 % and 95 %, respectively.

Of course, in applications the variance $\sigma^2(X)$ is usually even more difficult to compute than the mean value $E(X)$. It is therefore replaced by the "empirical variance":

$$s^2 = \frac{1}{N-1} \sum_{i=1}^{N} \left[X(\omega_i) - E(X_N)\right]^2 = \frac{1}{N-1} \left(\sum_{i=1}^{N} X^2(\omega_i) - \frac{1}{N} \left[\sum_{i=1}^{N} X(\omega_i) \right]^2 \right) \tag{5}$$

and one has also, under the assumptions made, for large sample size N

$$s^2 \to \sigma^2 \quad \text{and} \quad \sigma^2(I_N) \approx s_N^2 = \frac{1}{N} \; s^2 \; . \tag{6}$$

Hence, for large enough N, in the Gaussian based error estimates Eq. (4) σ can safely be replaced by s_N, at least for large sample size: $N \gtrsim 100$. In the opposite case: $N \lesssim 100$ "Student's t-distribution" should be employed in error analysis instead.

2 Random number generation

The Monte Carlo method rests on our ability to produce random numbers drawn from any particular probability distribution $F(x)$, or, if it exists, from the probability density function (pdf): $f(x)$, with $F(x) = \int_{-\infty}^{x} dt \; f(t)$.

Examples are wetting of a surface by rain (uniform distribution), radioactive decay (Poisson distribution) , or the distribution of velocities of molecules in a gas (Gaussian distribution). In general: the probability law must be known, either from theory, experiment or plausibility.

A theorem from measure theory states:

Theorem 1. *Each (probability) measure μ on $\mathbb{R}$ can be decomposed into a weighted sum $\mu = p_1\ \mu_c + p_2\ \mu_d + p_3\ \mu_a$ of three parts:*
Part 1 has a continuous distribution (with probability density $f(x)$)
Part 2 has a discrete distribution
Part 3 is a pathological ("singular with respect to μ") contribution

The third part is required for the abstract mathematical case only (general measurable spaces, $\sigma-$algebras, ...), but it does not occur in practical Monte Carlo applications. This means: for any distribution law arising in an application we can obtain random numbers in 2 steps: first a random decision (based on the two remaining weighting factors p_1, p_2) whether the continuous or the discrete distribution is to be sampled, and second then generating a random number from the chosen distribution μ_c or μ_d. We will show below that for both cases, continuous and discrete distributions, general procedures for random number generation exist, at least "in principle". We refer to the standard reference on the production of nonuniform random numbers [6]. (Also this book is out of print and freely available on the internet, see e.g.: http://www.eirene.de/Devroye.pdf) This book deals with the myriad number of ways to transform the uniform random numbers into anything else one might want. Also the first section, pages 1-193, of [8] is a very comprehensive introduction to random number generation.

2.1 Uniform random numbers

Uniform random numbers are the basis for generation of random numbers with all other distribution laws. A random variable is uniformly distributed on an interval [a, b], if the distribution density f is:

$$f(x) = 1/(b-a)\chi_{[a,b]}, \quad x\ \epsilon\ \mathbb{R} \tag{7}$$

with $\chi_{[a,\ b]} = 1$ if x is in the interval $[a,\ b]$ and $f(x) = 0$ elsewhere.

The classical method to generate uniform random numbers on [0, 1] is by so called "linear congruential random number generators", which are defined by the recursion:

$$\xi_{n+1} = [a\ \xi_n\ +\ b] \mod m \tag{8}$$

Here a is a "magic" multiplicand, m if often chosen to be the largest integer representable on the machine ($m = 2^{32}$, etc.), and b should be prime to m. Proofs for particular choices of (large) parameters a and m that the generator achieves the largest possible period of $m - 1$ different random numbers are quite cumbersome. Optimal parameter choices are typically found experimentally, see again [8]. The finite periodicity limits precision only in very large calculations, e.g. on modern massively parallel computing systems. A rather subtle issue is also "independence" of an entire sequence of random numbers (loc.cit.).

2.2 Non-uniform random numbers

As stated above we only need to consider discrete distributions and continuous distributions.

Finding random numbers with a given discrete distribution is trivial:

A discrete distribution, with k elementary outcomes labeled by natural numbers $\{0, 1, 2, \ldots, k\}$, can be given by:

$$P(X = i) = p_i \geq 0, \quad \sum_{i=0}^{k} p_i = 1 \quad \text{and} \quad F(i) = P(X \leq i) = \sum_{j=0}^{i} p_i \tag{9}$$

with $P(X = i)$ the probability of event i. F is the (cumulative) distribution. Let ξ be a uniform random number on [0, 1], then the random variable X with: $X = i$ if $F(i-1) < \xi \leq F(i)$ is distributed according to F.

Inversion method

The "inversion method" provides random samples z from a distribution F by converting uniform random numbers ξ. This is simply done by setting:

$$z := \min\{x | F(x) \geq \xi\} \sim F \tag{10}$$

If F is strictly monotonous, then $z = F^{-1}(\xi)$. For example: if $f(x)$ is the distribution density function (pdf) to be generated, then first find the cumulative function $F(x) = \int_{-\infty}^{x} dt\ f(t)$, pick a uniform random number ξ on [0, 1] and set $\xi = F(z)$ and finally invert this to find random number z, which is then distributed according to $f(x)$.

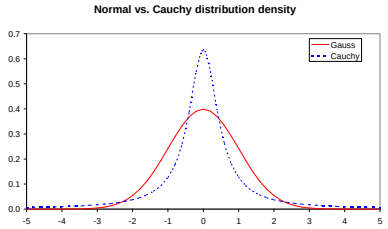
The same transformation rules as for any "density function" apply also for a pdf. Hence the general strategy is: try to transform a given pdf $f(x)$ to another distribution $\tilde{f}$, such that the inverse of the new cumulative distribution $\tilde{F}$ is explicitly known. Then apply the method of inversion and transform back. Figure 2 illustrates the method of inversion for the Normal (Gaussian) distribution:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x^2\right) \quad \text{and} \quad \Phi(x) = \int_{-\infty}^{x} dt\ \phi(t) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{x}{\sqrt{2}}\right)\right] \tag{11}$$

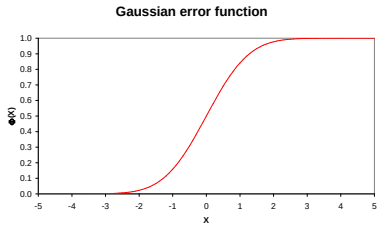
Unfortunately, the Gaussian error function $\operatorname{erf}(x)$ and hence $\Phi(x)$ cannot be inverted in closed form. We will show how to generate Gaussian random numbers, even without numerical inversion, further below.

From this procedure follows directly the natural and best format for storing (also multidimensional) tabulated data for random sampling in Monte Carlo applications:

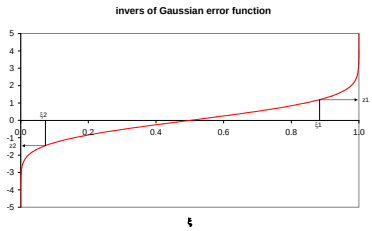
Form the inverse cumulative distribution function $F^{-1}(x)$ (i.e.: the quantile function) and store this for x uniformly spaced in [0, 1]. Then take ξ from a uniform distribution on [0, 1] and find $F^{-1}(\xi)$ by interpolation in this table.



(a) Normal vs. Cauchy distribution density



(b) Cumulated Gaussian distribution (error function)



(c) Inverse of cumulated Gaussian distribution

Fig. 2. *Top left:* Comparison of Cauchy (dotted line) and Normal distribution (solid line). *Top right:* Cumulative distribution function $\Phi(x)$ of Normal distribution, Eq. (11), *Right:* Inverse cumulative distribution of Normal distribution $\Phi^{-1}(\xi)$. Uniform random numbers ξ_1, ξ_2 (abscissa) are converted to random numbers z_1, z_2 from a Normal distribution (ordinate) as $z_i = \Phi^{-1}(\xi_i), \quad i = 1, 2.$

Rejection

Another general method for generating non-uniform random numbers is the rejection method (J.v. Neumann, 1947). This method is *always* applicable, although it may sometimes be rather inefficient. For distributions with finite support, i.e., $f(x) \neq 0$ only on a finite domain M (say, $M = [a,\ b]$), find the maximum c of $f(x)$, sample a random pair (ξ_1, ξ_2) with ξ_1 uniform on M and ξ_2 uniform on [0, c]. If $\xi_2 \leq f(\xi_1)$, accept ξ_1. Otherwise reject this pair and pick a new pair. Repeat this procedure until a pair is accepted. Clearly, the efficiency of this method (e.g. measured as average number of accepted random pairs to number of pairs produced) may be quite poor, in particular if the distribution $f(x)$ has sharp maxima.

A more general, and sometimes more efficient rejection method, working even on infinite sampling domains M, results if one finds a second distribution $g(x)$ and a numerical constant c such that $f(x) \leq c \cdot g(x)$. Again find a pair

(z_1, z_2) of random numbers, however with z_1 not sampled uniformly on M but from distribution $g(z)$ instead. z_2 is uniform on the interval $[0, c]$. The random variable z_1 is accepted if $z_2 \leq f(z_1)/g(z_1)$. Otherwise a new pair (z_1, z_2) is generated. See [5] for an important application in particle simulation.

Examples

Inversion Important examples in which the inversion method can be applied are, e.g., the *exponential distribution* (of the free flight length of radiation in matter), the *cosine distribution* of polar emission angles against surface normals, the *surfaces crossing Maxwellian flux distribution* $f(v_\perp) \propto v_\perp f_{Maxw}(v_\perp)$ of velocity components $v_\perp$ directed normal to the surface, for example of gas molecules with Maxwellian velocity distribution (f_{Maxw}) incident onto the container walls. We explicitly illustrate the inversion method here for the exponential (free flight length) distribution and for the Cauchy distribution:

The *exponential distribution* governs the free flight (length l) between collision events in all radiation shielding and kinetic transport simulations.

$$f_E(l)dl = \frac{1}{\lambda} \exp\left(-l/\lambda\right) dl \tag{12}$$

expresses the probability for an object traveling a ("free flight") distance l and then experiencing a "scattering event" in the interval $[l, l+dl]$. The parameter λ (dimension: length) is the so called "mean free path" in kinetic theory and is defined as the ratio of object speed v to collision frequency: $\lambda = v/n\langle\nu_c\rangle$, with n the density of scatterers, and $\langle\nu_c\rangle$ the collision rate coefficient (dimension: volume/time).

According to the inversion rule one samples the distance l by first integrating $f_E(l)$ and then equating that indefinite integral with a uniform random number ξ:

$$\xi = F_E(l) = \int_0^l \frac{1}{\lambda} \exp\left(-l'/\lambda\right) dl' = 1 - \exp(-l/\lambda) \tag{13}$$

and, hence, the result: $l = \lambda \ln(1-\xi)$. Since ξ and $1-\xi$ are both uniformly distributed, one may also use instead: $l = \lambda \ln(\xi)$ to generate a free flight distance.

By definition, the cumulative $F_E(l) = 1 - \exp(-l/\lambda)$ is the probability for at least one scattering event before a flight distance l, and hence $\exp(-l/\lambda)$ is the probability for a free flight of at least a distance l.

The *Cauchy distribution*, see Figure 2, top left, in physical applications also called "Lorentz" distribution, is an example of a distribution function that has no moments. It arises often in radiation transfer, e.g., as line-shapes of naturally- or Stark broadened lines or in other resonance phenomena.

$$f_C(x) = \frac{c}{\pi} \frac{1}{(x-b)^2 + c^2} \tag{14}$$

Here b is the median (line shift), and c is the half width at half maximum (HWHM). Generating random number with a Cauchy distribution is usually done by inversion. First transform to a standardized Cauchy, by $s = (x-b)/c$. The cumulative distribution is then given as

$$F_C(x) = \frac{1}{\pi} \int_{-\infty}^{x} \frac{1}{s^2+1}\, ds = \frac{1}{2} + \frac{1}{\pi} \arctan\left(\frac{x-b}{c}\right). \tag{15}$$

Therefore the random number $z = b + c \cdot \tan\{\pi(\xi - 1/2)\}$, with ξ a uniformly distributed random number on [0, 1], has a Cauchy(b,c) distribution.

The *cosine distribution* of a polar angle θ (measured against a surface normal) is a frequently applied distribution for the direction of emission of objects from a surface under equilibrium conditions (in physics often referred to as Lambert's cosine law). I.e. the emission is independent of an incident radiation, but rather is originating in the emitting body itself. The (bivariate) directional probability density $p(\theta, \phi) \propto \cos(\theta)$ for emission into solid angle element $d\Omega = \sin(\theta) d\theta d\phi$ of the half space is then given as:

$$f_{cos}(\theta, \phi) d\Omega = \frac{1}{2\pi} \cos(\theta) \sin(\theta) d\theta d\phi, \quad \text{for } \theta \in [0, \pi/2]\ \phi \in [0, 2\pi] \tag{16}$$

with the polar angle θ and the azimuthal angle ϕ measured against the surface normal. The factor $\frac{1}{2\pi} d\phi$ expresses the uniform distribution of ϕ. The cumulative distributions for θ is

$$F_{cos}(\theta) = \int_0^{\theta} d\theta' \cos(\theta') \sin(\theta') = 1 - \cos^2(\theta). \tag{17}$$

Therefore the random number $z = \cos(\theta) = \sqrt{1-\xi}$, with ξ a uniformly distributed random number on [0, 1], provides the desired information on the polar angle of emission.

The Box Müller Method for Gaussian random numbers Because the Gaussian error function cannot be inverted in closed form, the following combination of transformation, rejection and inversion method is typically applied:

Not one, but two independent normally distributed random numbers (z_1, z_2) are produced by first transforming random variables Z_1, Z_2 from cartesian to polar coordinates R, Φ. The angle Φ is then uniform in $[0, 2\pi]$. Only $\cos(\Phi)$ and $\sin(\Phi)$ are needed, and a rejection method (comparing a unit circle and a surrounding square) can be used for them. The variable R has, due to the Jacobian of the transformation, a Gaussian flux distribution (see above) rather than a Gaussian itself, and this can be directly generated by the Method of Inversion. Transforming back: $Z_1 = R \cdot \cos(\Phi)$ and $Z_2 = R \cdot \sin(\Phi)$ provides a pair of independent Gaussian random numbers.

3 Integration by Monte Carlo

Integration by Monte Carlo is a stochastic method for the deterministic problem of finding an integral, which in sufficiently complex high dimensional situations can be competitive or even superior to numerical methods.

Let's consider the source rate of particles (likewise, of momentum, heat, etc...) in a macroscopic system (e.g. a fluid flow), in which these particles (microscopic objects) are ruled by a kinetic, i.e. microscopic (Boltzmann) equation. Examples are: chemical sources (particle, momentum, energy) in plasma chemistry, or radiative heat source in case of radiation transfer theory.

Such terms then read:

$$I = \int_V dx\ g(x)f(x) := \int_V df\ g(x) \tag{18}$$

Here f is the *one particle distribution (density) function* $f(\boldsymbol{r}, \boldsymbol{v}, i, t)$ or $f(x)$, where the state x of the relevant phase-space may, e.g., be characterized by a position vector $\boldsymbol{r}$, a velocity vector $\boldsymbol{v}$, the time t, i.e. continuous variables, and further a discrete "chemical species index" i, also for example for internal quantal states. $g(x)$ is again some weighting function determined by the particular moment of interest.

In mathematical terms one would refer to this as Lebesgue-Stieltjes Integral of measurable function $g(x)$ with respect to (probability) measure defined by distribution density $f(x)$.

We will discuss Integration by Monte Carlo using the example from [2]: let the integration domain V be the unit interval [0, 1], $f(x)$ the uniform distribution on [0, 1] (i.e.: $f(x) = 1$ on [0, 1], and $f(x) = 0$ elsewhere) and $g(x) = \frac{e^x - 1}{e - 1}$. Clearly:

$$I = \int_0^1 \frac{e^x - 1}{e - 1}\ dx = [(e - 1) - 1]\frac{1}{e - 1} = 0.4180\ldots \tag{19}$$

We will now integrate this same function by Monte Carlo. Our first method does not require any theory, but instead, inspired by Buffon's needle experiment, we will just use pairs ξ_1, ξ_2 of independent uniform random numbers and compare the known area (the unit square $[0, 1] \times [0,\ 1]$) with the "unknown" area I, which is the area underneath function $g(x)$, in [0, 1]. I.e., we count a "hit" if the point defined by the pair of random numbers is under the curve $g(x)$, and a "miss" otherwise.

As can clearly be seen on Figure 3 the ratio of "hits" to total number of samples converges to the exact values of the integral, as expected, and also the statistical error, indicated as empirical standard deviation s_N, Eq. (5) scales with $1/\sqrt{N}$ as expected. In this trivial example we are dealing with a binomial distribution, the probability p for a "hit" is just the integral I itself: $p = I$. The statistical variance of the binomial distribution is known: $\sigma^2_{binom} = p(1 - p)$. Hence, if we repeat the experiment N times and average,

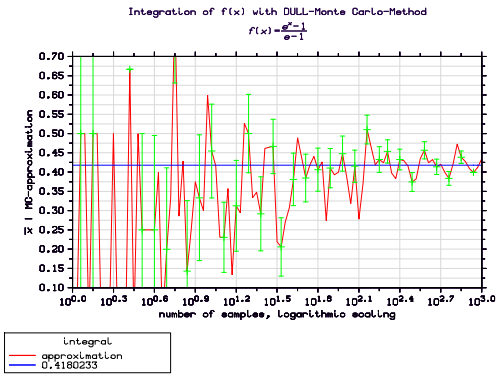


Fig. 3. Evaluating Integral of $e^x - 1/(e-1)$ on [0, 1], method: "hit or miss Monte Carlo", vs. number of trials (log scale). Blue horizontal line: exact value. Green vertical bars: empirical standard deviation s_N, (5).

then not only the mean value (the integral I) is known, but also the variance $\sigma_N^2 = p(1-p)/N$ and hence the standard deviation $\sigma_N = \sqrt{I(1-I)/N}$ and with it the confidence levels (error bars) are known exactly. In less trivial examples the variance is typically not known easily, but is itself then estimated from the random experiment ("empirical variance").

Of course such a Monte Carlo integration method is patently foolish. By this method we have, in principle, replaced the single integral over function g by a double integral over the area between abscissa and function $g(x)$.

The conventional text-book method ("crude Monte Carlo") can be obtained from this one by the observation that once the first random number ξ_1 of the pair is known, we do not have to rely upon ξ_2 to decide about counting zero or one. Given ξ_1, then a one will be counted with conditional probability p_{ξ_1}, and a zero else. $p_{\xi_1} = g(\xi_1)$. Hence instead we can use that (conditional) expected value p_{ξ_1} of the binomial distribution $b(1,\ p_{\xi_1})$ directly.

This is, admittedly, a quite obscure explanation for something really trivial. But it is also the underlying idea behind a powerful variance reducing Monte Carlo technique known under different names in different areas of application: "Conditional expectation estimator" (in neutron shielding), [4], "averaging transformation" (transfer theory, mainly in Russian literature), [7], or "energy partitioning method" in radiative heat transfer [9].

This method is opposite to "randomization": we have replaced a sampled result (zero or one) by its expectation value. In our particular example one may also say: we have carried out one of the two integrations analytically, conditional on the outcome ξ_1. The second random number is not needed at all in this particular trivial case but this not the relevant point. What is important also in general terms is that one (generally: some) of the two (generally: many) integrals has been done analytically, and only the remaining ones by random sampling. The general rule is: always try to do as many

"integrations" analytically or numerically and resort to Monte Carlo only for the rest. In particle transport theory this concept will lead to powerful hybrid methods combining information gained analytically (or numerically) and stochastically, bridging "continuously" the gap between stochastic and numerical methods. Sometimes, however, these resulting methods may loose their transparency.

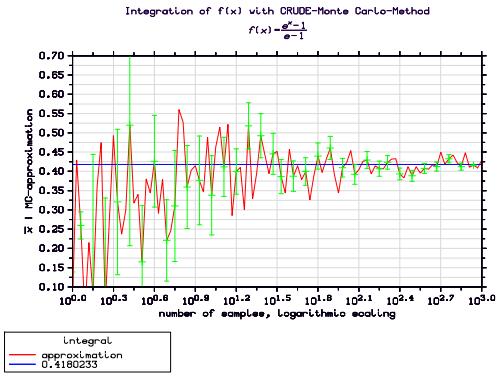


Fig. 4. Evaluating Integral of $e^x - 1/(e-1)$ on [0, 1], method: "crude Monte Carlo", red, blue and green data: as in Fig. 3

In this "crude" Monte Carlo integration the value of the integral I is obtained as estimated mean value (expectation value) of function $g(x)$ with respect to the uniform probability distribution $f(x)$ on [0, 1], $I = E_f(g)$, see remark after Definitions 2. Also indicated in Figure 4 is again the empirical standard deviation s_N, which, as expected, is significantly smaller than with the "hit or miss method" for any given number of trials.

Note that although this method has certainly a smaller statistical error per sample, the efficiency gain of one over the other method has also to account for the extra labor involved in evaluating the smoother estimator (which is hardly measurable in this trivial example chosen here).

In general Monte Carlo terminology one would refer to the uniform distribution $f(x)$ as the underlying stochastic law, according to which random samples X are produced. The random variable $g(X)$ is called estimator, score, or "response function".

Importance sampling

We are now in the position to explain the famous Monte Carlo concept of "importance sampling" for improving the statistical performance of Monte Carlo methods. Distinct from the "conditional expectation" technique discussed above, in which the statistical noise is reduced solely by modifying

(smoothing) the estimator $g(x)$, in "importance sampling" the underlying random variable (or random process) $f(x)$ is altered to another one, $\tilde{f}(x)$ in order to achieve variance reduction. A compensating weight correction factor is introduced in the estimator to maintain the same mean value $I = E_{\tilde{f}}(\tilde{g}(X))$:

$$I = \int_V g(x) \cdot f(x)dx = \int_V g(x)\frac{f(x)}{\tilde{f}(x)}\tilde{f}(x)dx = \int_V \tilde{g}(x)\tilde{f}(x)dx \ . \qquad (20)$$

Hence we have $\tilde{g}(x) = g(x)[f(x)/\tilde{f}(x)]$. The name of this method, "importance sampling" originates from the special techniques often used to find optimal biassing schemes (i.e.: $\tilde{f}(x)$) of the random process, in particular in transfer theory. A more general, but also somewhat imprecise terminology would refer to this concept as "non-analog" Monte Carlo, as compared to the "analog" Monte Carlo scheme. In the latter the underlying probability distribution law is directly taken from the application, whereas in the former one uses a different distribution, motivated by practical, economical or other reasons, and then applies statistical weights to compensate this.

As seen from Eq. (20), the value of I is independent of how the integrant is decomposed into a product of a probability density and a response function, but the variances, $\sigma^2_f(g)$ and $\sigma^2_{\tilde{f}}(\tilde{g})$, certainly can be very different.

Let's take, again, our example, to illustrate the concept:

In order to reduce the variance $\sigma^2_{\tilde{f}}(\tilde{g})$ of $\tilde{g}$ with respect to probability law $\tilde{f}$ we should try to make $\tilde{g}$ "as constant as possible" on [0, 1]. The Taylor expansion of our particular function $g(x)$ indicates that the ratio $\tilde{g}(x) = \frac{g(x)}{x}$ should be more constant than $g(x)$ itself. Hence we try $\tilde{f}(x) \propto x$, i.e., $\tilde{f}(x) = 2x$ so that $\tilde{f}(x)$ is normalized to one on [0, 1]. Our importance sampling

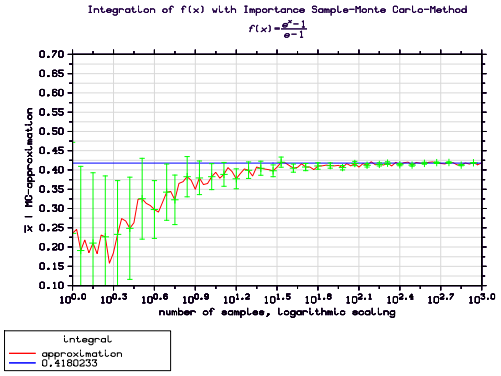


Fig. 5. Same integral as in Fig. 4, method: "importance sampling Monte Carlo"

procedure to evaluate I now proceeds as follows: Draw random numbers $\tilde{\xi}$ from $\tilde{f}(x)$. By the method of inversion, this is done by setting $\tilde{\xi} = \sqrt{\xi}$, with ξ

a uniform random number on [0, 1]. Then, again, form the arithmetic average of many (N) random variables $\tilde{g}(\tilde{\xi})$. Figure 5 shows the result of such an integration, again vs. N. Clearly the convergence is a) to the correct value, b) still only $\propto 1/\sqrt{(}N)$, but c) the the error bars s_N are much smaller than in both previously discussed Monte Carlo integration methods.

Again, it needs to be pointed out that the efficiency of the procedure is neither determined by the variance, not by N per CPU-time, but only by the figure of merit: "variance per CPU time". And hence, importance sampling, more generally: non-analog sampling, can go both ways in Monte Carlo. Its performance has to be assessed on a case by case basis.

Note: it is a general observation that in non-analog Monte Carlo schemes the error assessment simply based upon the empirical variance, and error bars obtained from the central limit theorem, can be less reliable than in analog simulations. Although the variance may be decreased by a clever importance sampling method, the "variance of the variance" may increase, thus invalidating conventional error bar estimates, see [10].

As in the case of "conditional expectation Monte Carlo" we can design an extreme case of importance sampling with zero statistical error after only one sample: Let us set $\tilde{f}(x) = g(x)/I$, hence: $\tilde{g}(x) = I = const.$ Monte Carlo integration proceeds by sampling from this distribution $\tilde{f}(x)$ which, in case of our particular example can be done by the rejection technique. Then, independent of the sampling, I is scored, with statistical error identical to zero (any trial will already provide the exact result). Unfortunately we needed the knowledge of the final result I already to design this perfect zero variance scheme.

δf Monte Carlo

Finally we use our simple integral to illustrate the concept of the δf Monte Carlo method, which is widely used in kinetic particle transport simulations. Starting point is the idea to split the unknown parameter into a large known "nearby" quantity and small unknown perturbation. In particle simulations this can also be the single particle distribution function $f(x)$ solving some kinetic equation or moments of this pdf. In near equilibrium situations we have

$$f(x) = f_{equil}(x) + \delta f(x)$$

with, for example, the Maxwellian equilibrium distribution f_{equil} and a small perturbation δf. It can then be advantageous to solve, by Monte Carlo sampling, only for δf rather than for the full distribution.

So let us consider our integral again, and write, accordingly, $I = I_0 + \delta I$ with I_0 the known part:

$$I_0 = \frac{1}{e-1}\int_0^1 dx \;\; g_0(x) = \frac{1}{e-1}\int_0^1 dx \;\; \left(x + \frac{1}{2}x^2\right) = \frac{2}{3}\frac{1}{e-1}$$

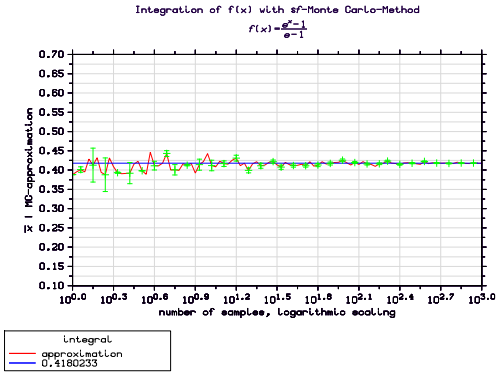


Fig. 6. Same integral as in Fig. 4, method: "δf Monte Carlo"

and δI the rest. Clearly,

$$\delta I = \int_0^1 dx \; \frac{e^x - 1 - x - x^2/2}{e - 1} \tag{21}$$

Figure 6 shows the result of the estimate for I, with I_0 known and δI evaluated by crude Monte Carlo. Clearly by eliminating a large, known, contribution to I the relative errors of the estimates for any given sample size N are greatly reduced as compared to previous methods.

This method is also related to the so called *correlated sampling technique*. Rather than trying to find a Monte Carlo estimate for δI to be added to the known part I_0, in this approach one would evaluate both nearby values I and I_0 by Monte Carlo techniques, but using the same (synchronized) sequences of random numbers. Both estimates are then positively correlated and the statistical precision of the Monte Carlo estimate for the difference δI can be substantially better than it would be in independent estimates of I and I_0 or of I alone. This method works best for estimating small incremental effects, even if both: I and I_0 are obtained only up to Monte Carlo precision. In the special case from above ("δf Monte Carlo), when one of the two, say I_0, is known exactly, one speaks typically of a "control variate technique" synonymously to "δf Monte Carlo", rather than using the more general notion "correlated sampling".

Multi-dimensional Integrals

The full advantage of Monte Carlo Integration, apart from its conceptual simplicity, lies in its superior performance in terms of both speed and precision (compared to numerical integration techniques) for higher dimensional integrals. So see this we generalize our 1D example to 2 dimensions (and similarly for arbitrarily many dimensions). Lets integrate in 2D: $g_2(x, y) = \frac{e^{xy}-1}{e-1}$, compare to (19). After some minor analytical work, one finds:

$$I_2 = \int_0^1 \int_0^1 \frac{e^{xy} - 1}{e - 1}\, dxdy = [(Ei(1) - \gamma) - 1]\frac{1}{e - 1} = 0.1850\ldots \qquad (22)$$

with Ei the exponential integral function and $\gamma = 0.57722\ldots$ being the Euler-Mascheroni constant.

Generalizing the Monte Carlo integrations to 2 (or more) dimensions means just using, per step, one (or more) additional random number (here: one more for the y coordinate value), and repeatedly evaluating g_2 with those sampled x, y values. For example, already with the "crude" Monte Carlo method we found experimentally, for $N = 100.000$: $I_{crude} = 0.18517\ldots$ and $s_N = 0.598\ldots\ 10^{-03}$.

Proceeding to higher dimensional integrals obviously remains trivial and precise with Monte Carlo, and by far not so with numerical or even analytical integration.

4 Summary

The purpose of this introduction was to show that a) random numbers can be generated from any given probability density distribution and b) Monte Carlo Methods can be regarded as stochastic (rather than numerical) procedures for integration. Monte Carlo consists of inventing a random game such that the expected value of a proper random variable is exactly equal to the parameter which is to be computed. Averaging over repeated independent Monte Carlo samples from that game converges (in the proper measure theoretical sense) to the desired solution.

The additional complication arising in many particle physics applications and in transfer theory is due to one fact only: Distinct from the material in this present chapter the sampling distribution $f(x)$ is sometimes not known explicitly. Instead it will be given only implicitly as solution of a, usually, very complicate equation (e.g.: the Boltzmann equation, the Fokker-Planck equation, etc.). We will see that this extra complication can be dealt with by sampling from certain stochastic processes ("generating particle trajectories"), rather than sampling from a given pdf, see [5]. But the rest: estimation of multi-dimensional integrals, the unbiased nature of the method, proof of convergence, error bars, variance reduction methods, remain essentially the same as in this present introduction.

References

1. H. Kalos and P. A. Whitlock, Monte Carlo Methods, Volume I: Basics, Wiley-Interscience Publications, John Wiley and Sons, New York, 1986.
2. J. M. Hammersley and D. C. Handscomb, Monte Carlo Methods, Chapman and Hall, London & New York, 1964.

3. R. Y. Rubenstein, Simulation and the Monte Carlo Method, Wiley Series in Probability and Mathematical Statistics, John Wiley and Sons, New York, 1981.
4. Spanier, J., Gelbard, E.M., Monte Carlo Principles and neutron transport problems, Addison Wesley Publication Company 1969
5. D. Reiter: The Monte Carlo Method for Particle Transport Problems, in: Computational Many-Particle Physics, Eds: H. Fehske, R. Schneider, A. Weiße, Chapter 5, Springer Lecture Notes in Physics 739 (2008)
6. L. Devroye, Non-Uniform Random Variate Generation, Springer-Verlag, Berlin, Heidelberg, New York, 1986.
7. G.A. Mikhailov, Optimization of Weighted Monte Carlo Methods, Springer Verlag, Berlin, Heidelberg, New York, 1992
8. D. E. Knuth, The Art of Computer Programming, Volume 2, Seminumerical Algorithms, Third Edition, Addison Wesley, Reading, MA, 1998.
9. A. Wang, M. F. Modest: JQSRT, A **104**, 288 (2007)
10. K. Noack, Ann. nucl. Energy, **18**, No. 6, 309 (1991)

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